

Attention Residuals Beyond Text: A Reproduction and Multimodal Extension

(working title — open to revision)

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Working analysis document — not a finished draft

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Purpose of this document. This is a working write-up of the observations and intuitions from my reproduction and multimodal extension of Attention Residuals (AttnRes, Moonshot AI). It is meant to share the current findings and my reasoning about *why* they occur, and to flag what is solid versus what is still in progress. It is not a polished paper draft. Intuitions are marked explicitly as hypotheses where they are not yet directly tested. All quantitative results below are over **3 paired seeds** (123, 456, 789) unless noted otherwise.

1 Summary of what AttnRes does (for context)

Standard residual connections update the hidden state as $h_l = h_{l-1} + f_{l-1}(h_{l-1})$. Unrolled, layer l receives a *uniformly* weighted sum of all prior layer outputs, and the residual-stream magnitude grows with depth, progressively diluting earlier contributions. AttnRes replaces this fixed sum with a learned, input-dependent softmax attention over the previous layer outputs: each layer carries a learned pseudo-query w_l , computes attention weights against the (RMSNorm'd) prior outputs, and takes a weighted sum. In the framing of the original paper, standard residuals are depth-wise *linear* attention and AttnRes is depth-wise *softmax* attention.

My implementation is *Full* AttnRes (unwindowed depth history; per-site learned queries on both the pre-attention and pre-MLP inputs plus a final depth readout; RMSNorm on keys; zero-initialised queries). I did not implement the paper's *Block* AttnRes variant. This distinction matters for one of the analyses below.

2 Results (3 paired seeds)

All comparisons hold the configuration fixed and differ *only* by `attnres.enabled`. For scaling runs: TinyStories, ctx512, 3000 steps, three model sizes (small ≈ 30 M, medium ≈ 51 M, large ≈ 92 M), 3 paired seeds (123, 456, 789) each. VLM runs: frozen SigLIP-Base/16@224 $\rightarrow$ linear connector $\rightarrow$ the large decoder, Flickr30K captioning loss, 6750 steps, 3 paired seeds. AttnRes adds a small number of depth-mixing parameters per size (+32K for large, +0.017%).

2.1 AttnRes gain scales monotonically with model size

Observation. The improvement is seed-stable at every size and strictly increases with scale: 7.6% $\rightarrow$ 13.3% $\rightarrow$ 15.1% perplexity reduction. At large scale the second-half-of-sequence loss reduction (0.185) is slightly larger than the overall loss reduction (0.164), suggesting the gain compounds with sequence position.

Intuition (hypothesis). The problem AttnRes solves — uniform-weight accumulation diluting earlier-layer contributions — compounds with depth. More layers means more accumulated dilution, so selective depth retrieval provides proportionally larger benefit. This is consistent with

Size	Baseline ppl	AttnRes ppl	Improvement	Val-loss reduction (paired)
Small ($\approx 30\text{M}$)	11.93 ± 0.05	11.02 ± 0.06	7.6%	0.081 ± 0.009
Medium ($\approx 51\text{M}$)	9.76 ± 0.06	8.46 ± 0.08	13.3%	0.142 ± 0.005
Large ($\approx 92\text{M}$)	8.49 ± 0.09	7.21 ± 0.03	15.1%	0.164 ± 0.007

Table 1: TinyStories, ctx512, 3000 steps, 3 paired seeds per size. The AttnRes advantage grows monotonically with model size. Error bars do not overlap at any size. AttnRes is consistently lower-variance across seeds than the baseline, most notably at large scale (± 0.03 vs ± 0.09).

the original paper’s $O(L)$ magnitude-growth motivation, extended here to show the empirical gain scales accordingly across independent model sizes. The slightly larger second-half gain is consistent with the hypothesis that longer-range composition benefits most from selective depth access, though this requires further investigation.

2.2 AttnRes transfers to a vision–language model

	Baseline	AttnRes	Reduction
Eval loss	2.873 ± 0.015	2.723 ± 0.009	0.150 ± 0.008 (paired)
Eval perplexity	17.69 ± 0.27	15.22 ± 0.14	13.9% relative

Table 2: VLM, Flickr30K captioning, 3 paired seeds. Per-seed eval-loss reductions: 0.158, 0.143, 0.149. AttnRes wins at every logged checkpoint, not only at the final step.

Observation. The original paper only evaluated text decoders. Swapping a standard-residual decoder for an AttnRes decoder inside a frozen-SigLIP VLM yields a $\sim 14\%$ captioning-perplexity improvement, seed-stable, with the same direction in a second hyperparameter setting. This is the core “push into a new setting” result.

Intuition (hypothesis). A captioning decoder must integrate two quite different representation types (projected visual features as a prefix, and text tokens). Learned depth-mixing lets later layers pull selectively from whichever earlier layers hold the relevant modality-specific information, rather than blending them with fixed uniform weights. I would treat this as a plausible mechanism, not an established one.

2.3 Modality-dependent depth utilisation (novel observation)

Because AttnRes exposes the depth-attention weights, I can measure where each token type allocates its depth-mixing weight. Splitting by vision-prefix tokens vs. language tokens:

(AttnRes VLM, 3 seeds)	Vision tokens	Language tokens
Embedding contribution	0.377 ± 0.013	0.215 ± 0.003
Depth-attention entropy	1.949 ± 0.022	2.136 ± 0.018

Table 3: Vision tokens place markedly more weight on the embedding/shallow source and attend more narrowly (lower entropy); language tokens distribute weight more broadly and toward deeper layers. Cleanly separated across all three seeds.

Observation. Vision-prefix tokens lean on shallow/embedding-level sources (embedding weight ≈ 0.38) and have lower attention entropy; language tokens distribute attention more broadly and deeper (embedding weight ≈ 0.21 , higher entropy). This is a pattern a text-only study could not have surfaced.

A related scaling observation (GPT-only, 3 seeds per size). The depth-utilization pattern also varies with model size. As model size increases, the embedding contribution *decreases* and entropy *increases*:

Size	Embedding contrib.	Entropy	Late contrib.
Small ($\approx 30\text{M}$)	0.281 ± 0.001	1.560 ± 0.016	0.456 ± 0.012
Medium ($\approx 51\text{M}$)	0.222 ± 0.002	1.822 ± 0.029	0.487 ± 0.026
Large ($\approx 92\text{M}$)	0.166 ± 0.000	2.124 ± 0.015	0.502 ± 0.020

Table 4: AttnRes GPT, ctx512/3000 steps, 3 seeds per size. Larger models rely less on the embedding as a dominant source and distribute depth-attention more broadly.

This suggests that as capacity grows, AttnRes learns more distributed depth-mixing strategies rather than collapsing to shallow retrieval.

Intuition (hypothesis, with a clean test). The vision encoder (SigLIP) is *frozen and pretrained*, so vision-prefix tokens enter the decoder as already-rich, high-level features — the useful signal is essentially present at the input (shallow) and the decoder does not need to rebuild it across depth. Language tokens enter as bare embeddings and must be composed into contextual meaning layer by layer, so their useful representation accumulates with depth. In short: *depth-attention puts weight where the useful information for that token already lives* — shallow for pre-processed vision, deep for from-scratch text. **Direct test:** unfreezing SigLIP (or using a weaker vision encoder) should push vision tokens to attend deeper if this asymmetry is the cause. I have not run this yet.

2.4 Gradient concentration and early-layer activation geometry

Using per-module norms logged throughout training (verified over the last 500 steps and across all three seeds, not a single-step snapshot):

(large/ctx512, steps 2500–3000, all seeds)	Baseline	AttnRes
Grad norm, block-0 attention	$\sim 86\text{--}117$	$\sim 6\text{--}8$
Grad norm, blocks 1–11 (attn)	$\sim 3\text{--}7$	$\sim 1\text{--}5$
Activation norm, block-0 attention	~ 1.65	~ 15
Global grad norm (clipped)	$\sim 0.7\text{--}0.8$	$\sim 0.7\text{--}0.8$

Table 5: The baseline concentrates a large *per-module* gradient norm in the first block’s attention ($\sim 10\text{--}15\times$ AttnRes), paired with collapsed first-block attention activations. AttnRes shows neither. Persistent across late training and all three seeds.

Observation. In the baseline, block-0 attention has *small forward activations* (≈ 1.65) but a *large backward gradient norm* (≈ 95), consistently across seeds and throughout late training. AttnRes shows normal-scale activations (≈ 15) and gradients (≈ 7) at the same site.

Important framing. This is a statement about gradient *distribution across layers*, not about training instability: the *global* (clipped) gradient norm is comparable ($\sim 0.7\text{--}0.8$) for both models.

So the precise claim is “baseline concentrates gradient in early-layer attention,” not “baseline gradients explode.”

Connection to the paper (partial replication). The original paper argues baseline residuals dump disproportionate gradient into early layers while AttnRes distributes more uniformly. The early-layer-concentration half *replicates* in my setting, but localised specifically to block-0 attention — baseline layers 1–11 are already fairly flat, so I would not claim “global uniformity,” only that AttnRes removes a pronounced first-block gradient/activation pathology.

3 Where the paper’s claims do *not* cleanly replicate

Activation-magnitude stabilisation. The paper reports that AttnRes yields more bounded hidden-state magnitudes across depth (mitigating PreNorm dilution). In my runs this is *not* clearly supported: last-layer activation norms for AttnRes are comparable or slightly higher than baseline, not lower. I see three candidate explanations, and I would value your read on which is most likely:

1. **Full vs. Block.** The paper’s stabilisation analysis is specifically about *Block* AttnRes, where block boundaries reset accumulation. I implemented *Full* AttnRes, which has no such reset mechanism, so the bounded-magnitude behaviour may simply not be expected here.
2. **Scale.** Their analysis is on a 48B model over 1.4T tokens; mine is $\sim 90\text{M}$ over short training. PreNorm magnitude growth is a deep-network phenomenon that compounds with depth, so at 12 layers there may be little for AttnRes to “fix.”
3. **Measurement.** I have been comparing endpoint (last-layer) norms; the paper’s claim is about the *depth profile* of norms. I have not yet plotted the full per-layer norm trajectory, which is what their claim is actually about.

The honest statement is therefore “I have not reproduced the exact setting their stabilisation claim applies to,” rather than “the claim is wrong.”

4 Open questions / where I would value your input

- **Full vs. Block.** Is it worth implementing Block AttnRes to test the stabilisation claim on its own terms, or is that out of scope for a first submission?
- **Vision-attends-shallow hypothesis.** Is the frozen-encoder asymmetry explanation the right framing, and is the unfreeze-SigLIP test worth running as confirmation, or is it a distraction from the core result?
- **Which thread to feature.** The strongest cards seem to be (a) the VLM transfer and (b) the modality-dependent depth utilisation. Should the paper centre on the multimodal extension, or give equal weight to the LM scaling behaviour?
- **Depth-profile analysis.** I plan to plot per-layer activation/gradient norms (baseline vs. AttnRes) to properly evaluate the stabilisation claim. Is there a specific diagnostic you would want to see?

5 Limitations (current)

- **Scale range.** Results span $\approx 30\text{--}92\text{M}$ parameters with short training (≤ 3000 steps text, 6750 VLM). Findings may not extrapolate to large pre-training scale.
- **VLM evaluation is captioning loss only.** No VQA, retrieval, CIDEr, or generation-quality metrics yet.
- **Full AttnRes only.** No Block variant, so claims tied specifically to Block AttnRes (e.g. the stabilisation result) are not directly testable here.

6 Method

AttnRes. Full depth AttnRes: unwindowed depth history, per-site learned queries on the pre-attention and pre-MLP inputs plus a final depth readout, RMSNorm on keys, zero-initialised queries. Standard causal self-attention is unchanged; AttnRes replaces only the depth-wise (cross-layer) aggregation. Toggled by a single config flag against an otherwise identical baseline.

GPT LM. TinyStories, GPT-2 tokenizer; $d_{\text{model}}=640$, 12 layers, 10 heads, $d_{\text{ff}}=2560$, context 512; 3000 steps; AdamW, lr $3 \times 10^{-4} \rightarrow 3 \times 10^{-5}$ cosine with 300-step warmup, $\beta_2=0.95$, weight decay 0.01, grad clip 1.0, fp16; effective batch 16 (batch 1×16 grad-accum).

VLM. Frozen SigLIP-Base/16@224 $\rightarrow$ linear connector $\rightarrow$ the same large decoder (baseline or AttnRes); visual features prepended as a prefix to text token embeddings. Flickr30K, captioning (masked-LM) loss on text logits only. 3 epochs, batch 8 (≈ 6750 steps), AdamW lr $1 \times 10^{-4} \rightarrow 1 \times 10^{-5}$, 100-step warmup, weight decay 0.01, grad clip 1.0; max caption length 128, 20,000 captions with a 90/10 split.

Reproducibility. All RNGs are seeded (`random`, `numpy`, `torch`, CUDA, and the data-loader generator). In our environment, repeated GPT runs with the same seed produced identical loss curves (empirically reproducible), although deterministic backend flags (e.g. cuDNN deterministic mode) were not explicitly forced. All reported comparisons use the same three seeds (123, 456, 789) for baseline and AttnRes, enabling paired comparison.